

DEHRADUN, India

WMO: 421110

Lat: 30.317N

Lon: 78.033E

Elev: 2238

StdP: 13.55

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	41.5	43.4	35.8	33.5	52.5	37.9	36.3	51.5	5.5	58.7	5.2	59.0	1.1	50	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	17.9	100.0	71.7	97.4	71.5	94.8	71.6	80.8	87.8	79.9	86.5	79.2	85.5	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.0	164.0	84.6	78.2	159.5	83.6	77.5	155.7	82.8	46.8	88.3	45.9	86.6	45.0	85.9	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	5.0	4.2	38.7	103.9	2.1	2.7	37.3	105.8	36.1	107.4	34.9	108.9	33.4	110.9		
(5)			37.7	83.1	2.0	1.6	36.2	84.3	35.0	85.2	33.9	86.1	32.4	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.2	55.8	60.6	69.1	77.7	83.0	83.9	80.8	80.1	78.6	72.9	65.4	58.5
	DBStd	10.31	3.57	4.26	5.28	4.66	3.94	4.36	2.41	2.21	2.56	3.60	3.17	3.38
	HDD50	2	2	0	0	0	0	0	0	0	0	0	0	0
	HDD65	680	287	133	21	1	0	0	0	0	0	1	33	204
	CDD50	8123	182	297	593	830	1024	1016	955	934	857	711	461	263
	CDD65	3324	1	10	149	380	559	566	490	469	407	247	45	1
	CDH74	33244	9	85	1292	4190	7093	7012	4564	3937	3140	1664	245	13
	CDH80	13885	0	7	386	2104	3953	3614	1357	1071	963	419	11	0
Wind	WSAvg	1.4	1.2	1.5	1.6	1.8	1.8	1.5	1.1	1.0	1.2	1.3	1.2	1.1
Precipitation	PrecAvg	83.4	2.1	2.3	1.7	1.2	2.0	6.8	25.3	26.7	11.8	2.0	0.3	0.6
	PrecMax	120.9	5.1	5.6	5.8	5.1	5.6	18.6	39.1	40.4	34.1	9.3	1.7	4.6
	PrecMin	52.5	0.2	0.0	0.0	0.0	0.0	0.0	6.7	15.3	3.1	0.5	0.1	0.0
	PrecStd	15.5	1.4	1.9	1.7	1.2	1.7	5.2	7.1	6.8	7.0	2.0	0.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	80.6	91.4	98.7	102.6	104.5	92.2	91.4	90.0	88.2	81.7	75.7
		MCWB	58.3	62.3	66.2	69.0	70.5	72.5	80.1	79.4	78.3	73.2	65.9	61.1
	2%	DB	70.3	76.7	87.3	95.8	99.9	101.0	90.0	89.0	88.4	86.0	79.0	72.6
		MCWB	56.3	60.7	66.0	69.4	70.6	73.3	79.2	79.3	77.4	72.1	64.5	59.1
	5%	DB	67.6	73.8	84.2	93.5	97.6	97.9	88.2	87.3	87.0	84.3	76.9	70.1
		MCWB	55.0	59.3	64.9	68.7	70.7	73.3	79.0	78.8	76.7	71.2	63.7	57.7
	10%	DB	65.0	71.2	81.1	90.8	95.3	94.4	86.6	85.6	85.5	82.5	74.6	67.6
		MCWB	54.1	57.9	63.8	67.5	70.7	74.5	78.6	78.2	76.2	70.2	62.6	56.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.7	64.6	70.9	76.4	79.4	81.4	81.9	81.7	81.0	77.2	68.9	62.9
		MCDB	71.5	76.3	85.1	93.0	93.9	91.5	88.8	88.0	87.5	84.2	77.9	73.2
	2%	WB	57.9	62.1	68.3	73.0	76.9	79.9	80.8	80.4	79.2	75.0	66.8	60.6
		MCDB	67.2	73.8	82.4	89.1	91.6	89.2	87.0	86.5	85.6	82.8	76.4	69.8
	5%	WB	56.3	60.4	66.3	70.7	75.2	78.9	80.0	79.5	78.2	73.4	65.1	58.8
		MCDB	65.0	71.3	80.6	87.6	90.3	88.4	85.8	85.2	84.4	81.2	74.5	67.6
	10%	WB	54.8	58.8	64.5	68.8	73.5	77.9	79.2	78.7	77.3	71.9	63.3	57.2
		MCDB	63.3	69.0	78.6	86.9	88.9	87.0	84.6	84.0	83.3	79.7	72.6	65.8
Mean Daily Temperature Range	5% DB	MDBR	21.2	21.0	23.1	24.7	23.5	17.9	10.8	10.9	14.0	20.3	22.0	22.3
		MCDBR	24.5	24.2	25.5	27.1	26.2	24.3	13.8	13.8	16.4	21.1	23.3	24.5
		MCWBR	11.9	11.1	10.0	10.5	9.9	8.7	6.3	6.6	7.9	10.2	11.4	12.2
	5% WB	MCDBR	22.0	22.1	23.6	24.4	21.7	15.8	12.2	12.0	14.3	17.1	21.8	22.7
		MCWBR	11.4	10.8	10.6	11.3	10.4	7.5	6.2	6.4	7.6	8.9	11.8	12.0
Clear-Sky Solar Irradiance	taub	0.403	0.429	0.440	0.500	0.582	0.656	0.648	0.559	0.493	0.474	0.422	0.397	
	taud	2.122	2.092	2.089	1.963	1.800	1.704	1.753	1.987	2.120	2.030	2.106	2.142	
	Ebn at Noon	255	261	267	255	235	217	218	237	249	243	246	250	
	Edh at Noon	41	47	50	59	69	76	72	56	48	49	42	39	
All-Sky Solar Radiation	RadAvg	1059	1337	1765	2068	2211	1926	1477	1442	1590	1622	1323	1096	
	RadStd	96	145	102	102	109	220	145	125	147	95	60	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+2.03	N/A	+0.90	+0.77	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	+0.36	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page